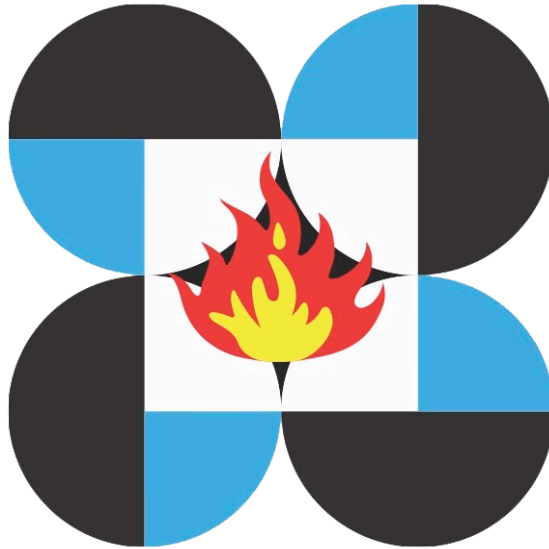


DEVELOPMENT OF A LABORATORY INVENTORY AND BORROWING  
SYSTEM FOR THE BIOLOGY LABORATORY IN THE PHILIPPINE SCIENCE  
HIGH SCHOOL - CAR CAMPUS IN BAGUIO CITY

This research proposal is submitted to the  
PHILIPPINE SCIENCE HIGH SCHOOL - CAR CAMPUS IN BAGUIO CITY  
in partial fulfillment of the course requirements in Research 2.



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## Introduction

### Background of the Study

Philippine Science High School system, often referred to as the PSHS system, gives focus to Science education which need not just theoretical knowledge but require practical and experimental work and application. The main venue for this experimental work is the laboratory which serves as an important foundation for student learning. Hands-on activities allow students to apply concepts learned from teachers in classroom discussions. This also allows students to test their knowledge and observe science in the real world. However, the effectiveness of these laboratories rely on a number of factors and this study aims to deal with one of them, which is Laboratory Inventory Management, which is described as the supervision of supply flow, all the way from the acquisition of the raw materials to their consumption, as well as the maintenance of equipment, which includes but is not limited to specialized physics equipment, high-value biological apparatuses, and volatile chemical reagents. Laboratories that are funded adequately, along with organized science laboratory management and proper administration, may improve student outcomes according to Neng Gustini and Wulandari ([2020, as cited in Santoso et al, 2024](#)).

Empirical studies have been conducted often underscore the importance of management of laboratories. A study conducted in La Union, Philippines, shows that incorrect/improper inventory and safety procedures were a detriment to proper resource utilization and learning opportunities. [Nuesca \(2025\)](#). However, the opposite was found in maintained laboratories described as, well-equipped, reliably stocked, properly maintained. With proper laboratories showing higher student achievement ([Mukti, 2020](#)) and displaying higher participation, enhanced critical thinking, and improved scientific reasoning ([Lazaro, 2025](#)).

Furthermore, laboratory inventory management systems (LIMS) that rely on paper logs, spreadsheets, and non-automated methods are recognized as inadequate for current laboratory operations. Studies show that these types of LIMS vastly underperform in current laboratory settings, showing it to be inefficient, prone to errors and unsustainable as it was poor at record-keeping as well as generating inaccuracies in stock data and being difficult to maintain accurate inventory levels without the use of digital tracking. This leads to problems such as wasted resources and disruptions in workflows ([Boche et al., 2022](#)). Specific researches that focus on laboratory environments demonstrate that manual operations make it difficult to track expiry dates, sample usage, and equipment availability, contributing to inconsistencies and allowing for more human error in inventory oversight ([Lazaro, 2022](#)). Additionally, Traditional manual logs struggle with real-time visibility and centralized control, resulting in redundant orders, stockouts, and resource misallocation, issues that computerized systems and LIMS are designed to reduce through automation, barcoding, and digital tracking ([Madamidola et al., 2024](#)). These findings align with broader research showing that manual systems lack the accuracy and scalability needed for effective inventory management and underline the value of adopting automated solutions to improve inventory accuracy, reduce human error, and enhance overall laboratory efficiency ([Boche et al., 2022](#)). Several studies have shown that problems exist with the current manual inventory system management, often regarding the reliability and accuracy of data. Additionally, research has indicated that developing digitized inventory systems significantly improves efficiency and reliability in recording data, as shown in the study of [Elec et al. \(2025\)](#), where a barcode-based inventory system greatly improved the existing manual inventory process. Other studies, such as [Ramos-Miller & Pacheco \(2023\)](#), have also shown that

web-based inventory systems not only enhance efficiency by being able to track real-time data but also improve accessibility, allowing the integration of borrowing systems and creating a more convenient transaction process. These support the need for the implementation of a digitized inventory and borrowing system for school laboratories to improve the reliability of data, efficiency of transactions and tracking, and accessibility for both students and teachers.

A cycle forms due to the combination of these inefficiencies, where outdated tracking methods reduce not only inventory visibility, but also hinder resource allocation and create bottlenecks in laboratory workflow. Manual systems are prone to loss, mishandling, and errors, which make them inaccessible, inaccurate, or not-up-to-date ([Mawire, 2022](#)). Meanwhile [Herlina \(2025\)](#), states that laboratories relying on manual systems experience delays and chemical waste due to unaccounted inventory.

Previous research also highlights challenges in secondary school laboratory management, such as limited equipment, weak administrative systems, and understaffing ([Rini et al., 2024](#)). Few studies examine the implementation of digital inventory tools specifically for school laboratories. Most existing research focuses on higher education, such as university LIMS ([Cortez, Meneses & Tan, 2024](#)), leaving a gap in understanding how automated inventory systems could enhance accountability, operational efficiency, and resource accessibility in specialized institutions like the Philippine Science High School Cordillera Administrative Campus in Baguio City (PSHS-CARC in Baguio City).

At present, manual inventory and borrowing processes continue to be used in the PSHS-CARC in Baguio City for laboratory equipment and materials. This

highlights the issues with the manual laboratory system and the need for a more efficient, reliable, and accessible inventory and borrowing system.

## **Objectives of the Study**

### ***General Objective***

This research study aims to develop a web-based inventory and borrowing system for the PSHS-CARC in Baguio City biology laboratory as of school year 2025-2027, to see whether it will enhance the efficiency of the current laboratory inventory and borrowing system

### ***Specific Objectives***

Specifically, this research aims to:

1. Design and develop a web-based laboratory inventory and borrowing system that is easily accessible for students, teachers, and laboratory personnel through different web browsers having a functional and user-friendly graphical user interface.
2. Implement an organized inventory management and borrowing system that classifies laboratory equipment using filters, allows users to check the availability of materials, and easily requests the needed materials.
3. Assess the developed system in comparison to the currently used inventory system in terms of transparency, reliability, and user satisfaction among students, teachers, and laboratory personnel using a questionnaire.

## **Significance of the Study**

This study allows for the encouragement of web-based or automated developments in school settings, contributing to the limited empirical research on the

implementation of automated laboratory inventory systems. While previous studies have explored laboratory management practices and their effect on student learning, few examine how LIMS can enhance real-time tracking, reduce human error, and improve accountability in science laboratories. This study benefits laboratory personnel, teachers, and students. For laboratory personnel, record-keeping would be more accurate and would reduce errors as well as manual work. Serving as a streamlined inventory management system, this helps them focus on maintaining equipment quality, preventing stockouts, and ensuring a safer laboratory environment. Furthermore, a developed automatic system would help them easily know which materials need to be restocked and returned. On the other hand, teachers and students gain access to the necessary information about the laboratory materials, such as their availability and the quantity of materials available in real-time, without having to ask the laboratory personnel personally.

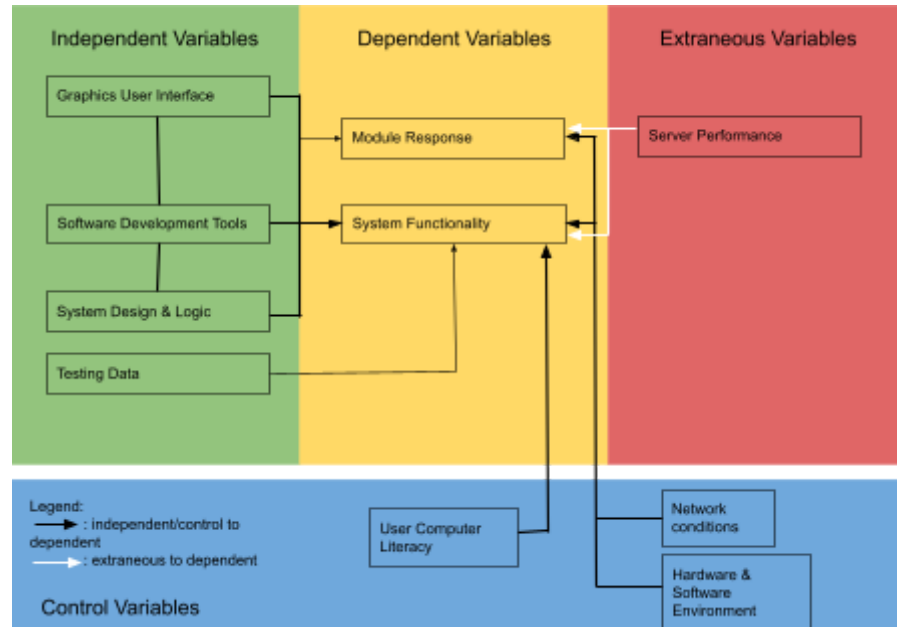
### **Scope and Delimitation of the Study**

This study focuses on the design, development, and initial deployment of a web-based laboratory inventory system specifically for the biology laboratory in PSHS-CARC in Baguio City. The project uses the Agile development methodology, focusing on delivering a functional tool within the project timeline. The tool aims to create a database that allows for comprehensive item management as well as expiration date monitoring and generation of stock level and usage history reports. The system is designed to digitize manual inventory processes. The findings of the study and the developed system itself are directly applicable only to the PSHS-CARC's specific data structure and laboratory environment. This study will not take into consideration constraints in server capacity, coding time, and development resources that limit the complexity of the system.

## Conceptual / Theoretical Framework

**Figure 1**

*Schematic Diagram of the Theoretical/Conceptual Framework*



This study considers the following core variables:

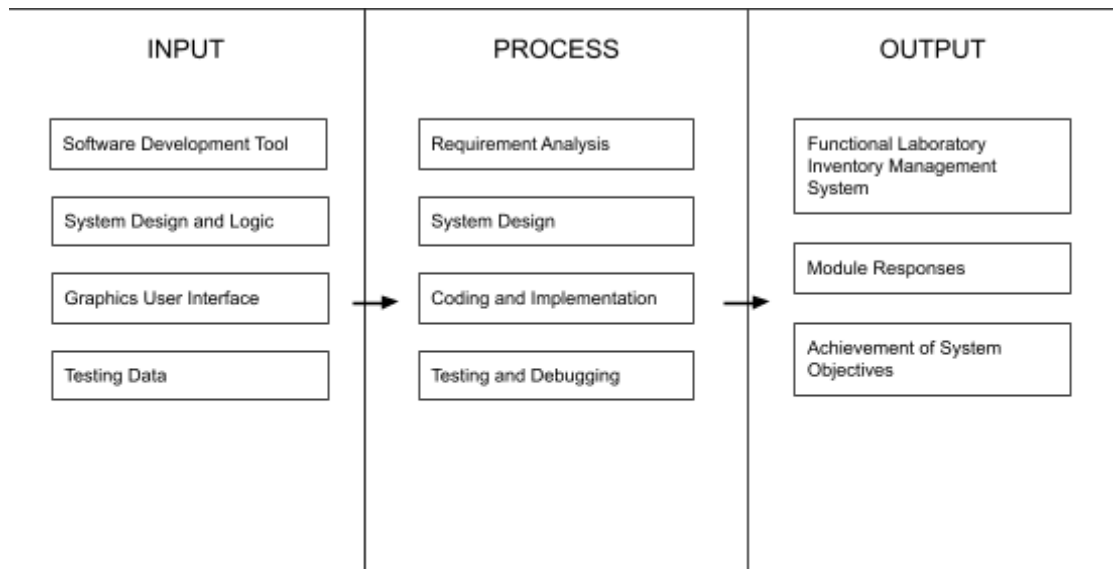
Independent Variables: Graphics User Interface, Software Development tools, System Design & Logic, Testing data.

Dependent Variables: System Functionality, Module response

Control Variables: Hardware Environment, Software Environment, Network Conditions, User Computer Literacy

**Figure 2**

*IPO model*



The conceptual framework the researchers will be using is best represented using the Input-Process-Output, otherwise known as the IPO model.

INPUT: The resources and technical decisions that define the system creation.

a) Software Development Tool:

- This is the specific programming language, framework, and database that the research chose to use (e.g., PHP, JavaScript for language, Laravel for framework, MySQL for database)

b) System Design & Logic

- The specific architecture and algorithms used in the database research define

c) Graphics User Interface (GUI)

- The front-end design, layouts, and wireframes chosen for user interaction

d) Testing Data

- Mock data the researchers use to simulate the inventory items, users and transactions that students, teachers and the likes do in the Philippine Science High School Laboratories.

PROCESS:

This stage involves the actual work done to show the output from the researchers' input. Since this is a developmental study, it is typical to use the Software Development Life Cycle. First, the researchers define the requirements for creating and manufacturing the database schema. Second, coding and implementation, where the researchers write the back-end logic and front-end GUI using the aforementioned Software Development Tools. Then, the Internal testing and debugging, where there will be final checks and debugging to ensure that the system's GUI and logic function as planned.

OUTPUT:

Lastly is the output, which includes the system functionality and the module response, which measures the successful execution of core features and the successful implementation of the specific objectives of the website, and the module response, which successfully shows the result from the system's core features and if each module responds accordingly to how it is meant to.

### **Review of Related Literature**

Inventory systems are more commonly developed in companies under retail and production, where they serve as a tracker and monitor for the arrival and exit of goods, ensuring that there is an accurate amount of inventory at any specific time period. Nowadays, the process of inventory systems has been widely automated and is highly encouraged for every organization due to more efficient capabilities, such as taking less time and effort. Although these systems have been widely implemented, it are not as common for school institutions as they lean more towards quality instruction ([Tapado & Delluza, 2016](#); [Madamidola et al., 2024](#)).

Inventory systems are often either manual or automated, with both having their advantages and disadvantages. According to [Noreen and Shaukat \(2015\)](#), manual systems are usually written on paper, tending to be extensive and require more time and effort compared to automated systems, and heavily rely on how one deems accuracy as important. On the other hand, automated inventory systems are made to decrease the load for personnel, being less time-consuming and laborious, while also reducing human errors ([Atieh et al., 2016](#)). However, as cited in [Tapado & Delluza \(2016\)](#), accuracy is not an advantage since automated systems still require the manual process of counting goods and may have fabricated results. For laboratories, automated systems are highly encouraged, supported by the article of [Hashim and Arifin \(2013\)](#), which states that in manual systems, it would be time-consuming for technicians to go over their files whenever there is a need to check equipment and track down the status of equipment.

Regardless of its type, an inventory system consists of components that allow industries to be able to monitor and track their inventories efficiently. According to [Raj \(2024\)](#), these components include demand forecasting, to see which items are in

demand and needs to be stocked to be able to keep up with that demand; inventory tracking, where the increase and decrease of stocks are being monitored in real time to ensure accuracy in the number of goods; reorder points and safety stock, which determines when to reorder items before they run out of stock depending on its demand; and inventory turnover which determines the performance of a product in sales. These components ensure that inventory systems remain accurate in stock tracking, reduce product waste, and are efficient in managing inventory. As such, these components can be adapted to track laboratory equipment inventory and monitor the borrowing and usage of such items, most especially in school laboratories, to inform both students and teachers about what materials and when they can borrow them.

Several researches highlight the issues that are being faced when using a manual inventory system with one being the study of [Mawire \(2022\)](#), where its analysis on the usage of paper-based or manual inventory systems at the Nyanga District Hospital highlighted issues such as the loss of and the mishandling of records, its unsystematic and time-consuming aspect, and are inaccessible in a way that stakeholders do not know the availability of items. Additionally, other research articles back up these claims, stating that manual systems are susceptible to data loss, human errors, and theft, which results in discrepancies in data and inaccuracy that will require more work to fix [Bhandare et al., \(2025\)](#). In conclusion, manual inventory systems are highly inefficient and require more work to be done than needed, and are also greatly susceptible to various issues, which can be lessened when automated systems are used instead.

Laboratories have started to switch from a manual inventory system to a more automated one, although there are still a lot that remain with the traditional

paper-based system. The inefficiencies of manual systems also apply to laboratory systems, as found in the study of [Herlina \(2025\)](#), which found that laboratories that incorporate manual inventory systems using check sheets often find discrepancies in their inventory and have difficulty in finding needed chemicals, leading to more chemical waste because the chemicals that are not accounted for and are only stored until they expire. Additionally, laboratories also experience errors and delays when manually tracking their resources ([Adekundle et al., 2024](#)). These problems aren't suitable for school laboratories where teachers and students often borrow apparatus, and thus, need an accurate, real-time update of which equipment and chemicals are available for use. In relation to this, several studies have also shared claims that a more systematic or automated approach is better, including the study of [Ramos-Miller & Pacheco \(2023\)](#), where a web-based management system was developed and later found to have improved efficiency and time consumed in managing inventory.

With the widespread use of barcode technology, quick response (QR) codes, and radio frequency identification (RFID), their limitations have surfaced. In the study of [Elyasizadeh \(2016\)](#), a comparison was made between barcode and RFID systems, indicating that barcodes, similarly to QR codes, are an inexpensive alternative to manual systems, but are susceptible to fading and tearing. On the other hand, RFID systems are better since they allow automatic tracking without needing to be in the line of sight and also reduce the threat of deception. Although this is true, it is costly and highly technical.

While automated systems reduce manual errors, limitations still exist in technologies such as barcodes and QR codes, suggesting the need for a more accessible solution like web-based systems. [Gardella et al. \(2024\)](#) developed a web-based inventory system for a university to counteract limitations such as

accessibility and item tracking. It was found that the expanded accessibility allowed for faster communication between staff and students and has overall greatly improved the efficiency and organization of their inventory. Although web-based technologies would still need manual checking of inventory systems, thus not counteracting human error, they allow for a wider range of accessibility, which, in the case of school laboratories, is more important due to the continued use of apparatus by both students and teachers.

Furthermore, [Tanaman et al. \(2023\)](#) also developed a web-based inventory system for a small business to counteract the issues of manual inventory, such as inaccuracy, inefficiency, and lack of real-time data. This system allowed for the inventory processes to be done online, digitizing forms, and provided real-time data tracking of inventory. It was found to have greatly increased the efficiency, reliability, and security. This supports how web-based systems are a great alternative to manual systems, given the setbacks of other systems, such as QR and barcode.

In the study of [Elec et al. \(2025\)](#), they developed a barcode-based inventory and borrowing system for the Philippine State College of Aeronautics due to similar problems with the manual inventory systems. The system that the researchers proposed is a Python system wherein, by scanning the barcode on the intended item to borrow and the identification of the borrower, information will be saved about the tool, its borrowing time, date, and status, and the borrower's information. This means that each of the tools and individual borrowers will have their own barcode. Furthermore, their system tracks the transactions made, also allowing the students, teaching staff, and non-teaching staff to see which tools are available and who they are with. Although deemed highly reliable and efficient, it raised some security issues, such as the data privacy of the borrowers.

**Elaboration of the Research Problem**

Automated inventory systems have been widely used in various organizations, but rarely in school institutions, most especially in the laboratories. Most research on automated inventory systems is catered towards actual laboratories, where very few include those of schools. In PSHS-CARC in Baguio City, this is also evident, wherein laboratories continue to use the manual checking of inventories, and both teachers and students need to ask laboratory personnel for the availability of apparatus personally. Currently, there are no studies on improving the laboratory management and borrowing system at PSHS-CARC in Baguio City, highlighting the need for an automated inventory system at the school. This study aims to develop an automated inventory system in the form of a website where the laboratory personnel of the biology laboratory in PSHS-CARC in Baguio City can input the availability of their equipment, and allow the teaching staff and students to personally view which of those are available and which are not. Additionally, the borrowing system will be incorporated into this system, enabling an easier and faster way to borrow items from the laboratory.

## **Methodology**

### **Research Design**

The research study will use a developmental design as the study focuses on creating an inventory system. A developmental design aligns with our objective of enhancing the current inventory system for the biology laboratory of PSHS-CARC in Baguio City, supporting an iterative process of creating and continuously improving a system to facilitate inventory management ([Richey & Klein, 2014](#)). The study focuses on the relationship between the design factors (Graphical user interface (GUI), Software development tools, System design and logic, and Testing data) and functionality, module response, and user response. While focusing on these relationships, the hardware and software environments, network conditions, and user computer literacy are treated as control conditions.

The study adopts an Agile-based developmental approach, where system components are developed, tested, evaluated, and refined iteratively. This approach supports continuous feedback from system functionality, module response, and user response throughout the development life cycle.

The developmental design that will be used by this study guides the development of an inventory system done iteratively and sequentially to facilitate its core functionalities. The independent variables dictate the structure of the interface on how the system's logic processes different laboratory resources and how the software manages data flow. Each design choice will be evaluated based on its effects on system functionality, module response, and how the system creates, reads, updates, and deletes data on laboratory resources.

The study evaluates each module on the execution of the intended logic, including the verification of functionalities such as the filters based on specific categories, availability tracking, and dissemination of borrowing rules and role-specific permissions. These are the components of the dependent variables and are the basis for whether or not the enhanced system meets its intended functional criteria.

The study compares user responses of students, teachers, and laboratory personnel using the questionnaire provided regarding the current inventory management system and the enhanced inventory system under the characteristics of transparency, reliability, and satisfaction.

The study will implement a structured data management plan to ensure security, confidentiality, and proper handling of user information within the inventory and borrowing system. The system will collect and store only minimal personal data that is necessary for authentication and accountability, specifically, the user's email address, full name, and password, without the collection of any identifiers such as contact numbers, addresses, or identification numbers. It will follow the principle of data minimization, wherein only information strictly required for system functionality will be processed.

All of the user credentials will be protected using Django's built-in authentication and password hashing system, wherein the system will be implementing role-based access control to ensure that students and teachers can only view inventory data and submit borrowing requests, while laboratory personnel can update inventory records and approve transactions so that only the designated administrators will have system-level privileges, and direct database access will be restricted to authorized personnel. Data transmission between users and the server will

be protected through HTTPS encryption. User activity logs will record system actions such as borrowing requests and inventory updates to maintain accountability.

Survey responses that are collected during the pre and post-implementation phases will be anonymized before analysis, and personal data will be retained only for as long as it's necessary for system operations and research evaluation. Research-related identifiable data will be deleted in accordance with school policy. Regular database backups will be scheduled to ensure recovery when needed.

The developed system will follow the principles of the Data Privacy Act of 2012, which states that users will be informed that their email, username, and real name are collected for authentication, borrowing accountability, and system management. This ensures that user data will remain protected against unauthorized access, misuse, and data breaches.

### **Agile Development Cycle**

The process of developing the enhanced inventory system follows a simplified software development life cycle implemented in iterative increments. The researchers first define the functional and data requirements of the system, the database schema, core modules, and user roles. The researchers will then implement the back-end logic and the front-end interface before the enhanced system undergoes testing and debugging to ensure that the functionalities of the system operate as intended.

This process gives the researchers the system functionality and module responses that indicate if system features operate correctly and if each module follows its intended design. The process reverts back to earlier stages for refinement to rectify any issues identified earlier in the process.

### **Samples and Locale of the Study**

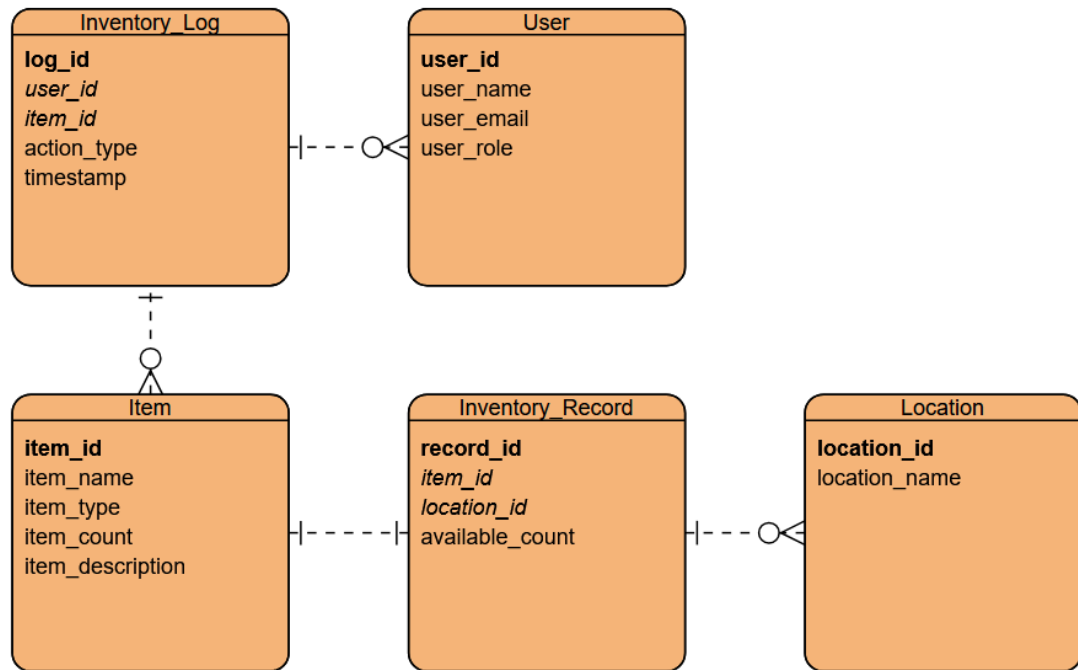
The study will be conducted in PSHS-CARC in Baguio City, with the inventory system being catered toward the biology laboratory and for the students, teachers, and lab personnel who use the laboratory. The inventory system that the study will be developing will contain records of the equipment and materials found in the laboratory.

For data collection after the implementation of the enhanced system, the target population for perception-based metrics (Transparency, Satisfaction, and Reliability) will be the students, teachers, and lab personnel who have used the laboratory and the inventory system. The study invites all that meet this criteria to participate in the survey that will be used to assess these metrics. Two surveys will be conducted, one for the current inventory system and the other for the enhanced inventory system.

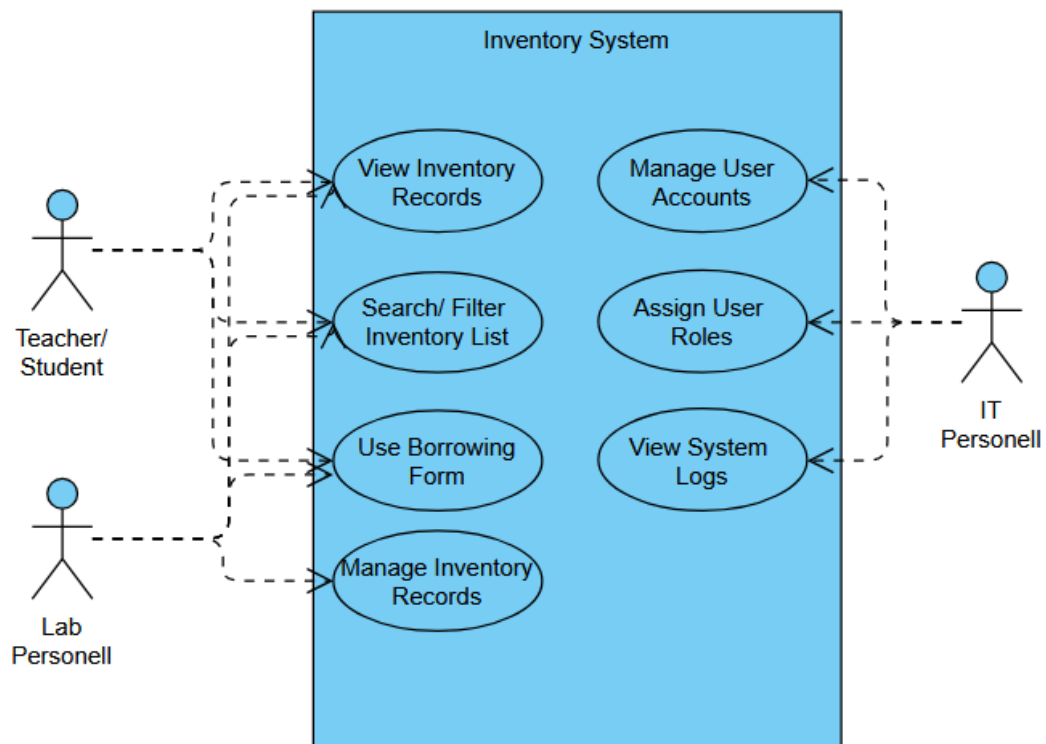
### **System Modeling and Design Artifacts**

#### **Figure 3**

*Entity Relationship Model (ERM) of the Enhanced Inventory System*

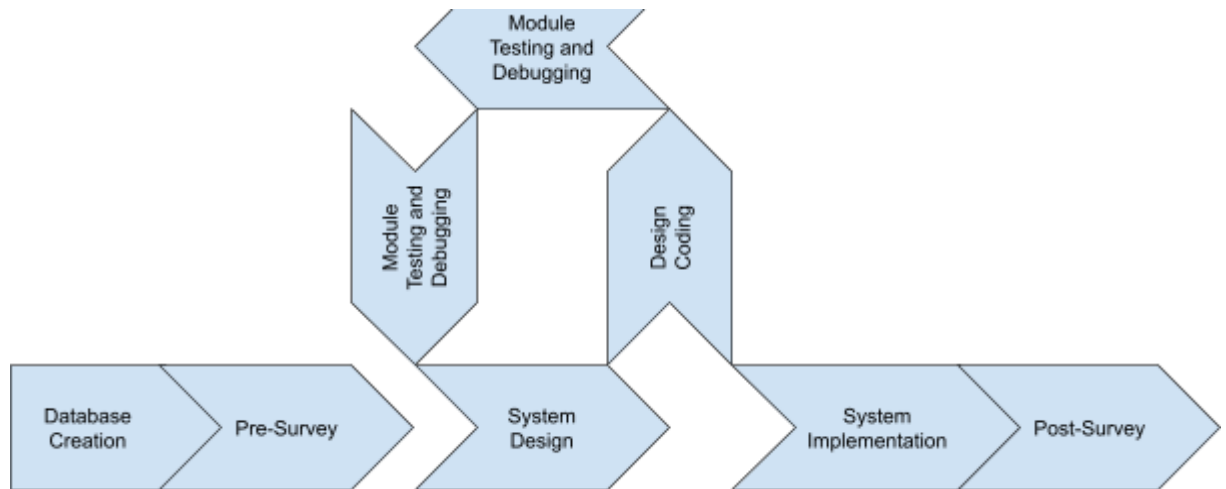
**Figure 4**

*Use Case Diagram of the Enhanced Inventory System*



**Data Gathering Procedure**

**Figure 5**

*Process Flow of the Development Study***Database Creation**

At this stage, the researchers collected inventory records from the laboratory until they had a complete list of the laboratory materials and equipment, along with counts and locations. This stage also had the researchers organize the inventory lists in a way that was presented in the enhanced inventory system. Each item will be organized based on its location, type (Material or Equipment), name, and count.

Specifically, a relational database management system (RDBMS) utilizing SQL (Structured Query Language) will be used. RDBMS is a type of database that organizes data into different tables, where each table is connected by relationships within the data using primary and foreign keys. Primary keys are assigned to rows in the table to identify a record, while foreign keys create references between tables ([Jemal, 2020](#)).

**Pre-Survey**

To have metrics to use for comparison of the current and enhanced system, the researchers have conducted a survey parallel to the database creation stage using a

questionnaire utilizing a five-point Likert scale; statements would be rated by the user from 1 (Strongly Disagree) to 5 (Strongly Agree).

This questionnaire consists of statements organized under the perception-based metrics (Transparency, Reliability, Satisfaction, Ease of Use). This survey has been distributed through the use of the school's emails and will only utilize the responses coming from users who meet the criteria of being a student, teacher, or laboratory personnel who have used the laboratory and the inventory system. The questionnaire used is provided below the data analysis.

### **System Design**

After establishing an RDBMS, the researchers will design a graphical user interface (GUI) for easy navigation for users when interacting with the RDBMS. To allow this easy navigation, the GUI must give users the ability to search, apply filters, view item availability and stock counts, and for lab personnel to be able to update the database. These functionalities aside, the GUI must be able to let the user, regardless of their technical background, access and utilize the RDBMS to promote accessibility ([Srivastava et al., 2020](#)).

The study will be utilizing Django, a Python web framework, to design and implement the frontend for the enhanced inventory system. Django provides a robust and scalable framework that can be utilized for CRUD functionality and operates well with SQL ([Benson-Emenike et al., 2023](#)). This makes Django suitable for designing inventory systems that implement the specific objectives.

### **Design Coding**

The finished GUI will then have its features implemented via structured query language (SQL) to allow CRUD (Create, Read, Update, and Delete data) functionality

in the RDBMS. SQL is the standard language used to communicate with the RDBMS as the nature of SQL being a declarative language allows the database engine to optimize queries internally for speed and efficiency ([Jemal, 2020](#)). SQL communicates with the RDBMS by giving instructions on what to do, but does not explicitly tell how.

After communication between the GUI and RDBMS is implemented, features not strictly related to CRUD functionality will be implemented. These features, such as logging in to differentiate between lab and non-lab personnel and background features such as records of actions, uptime, and error events, are ones that operate without active user input.

### **Module Testing and Debugging**

This stage will have the researchers conduct tests on the GUI simulating the system being used by students, teachers, and lab personnel. This will allow any bugs or issues from the design coding to be identified and rectified ([Madamidola et al., 2024](#)).

When such bugs or issues are revealed, the researchers will return to the design coding stage to make iterations until a satisfactory code design is created. Once a functional and consistent design is created, the researchers will proceed to the implementation of the enhanced inventory system. Each module will be tested against their intended purpose as defined by the technical objectives of the system.

Module response will be evaluated in terms of if each module correctly executes its expected functions, responds without error, returns correct information based on input, and if user restrictions are applied correctly based on the user role.

### **System Implementation**

At this stage, the satisfactory design will be implemented in PSHS-CARC in Baguio City, with access being granted to the intended users, being the lab personnel, teachers, and students. Different account types will be assigned to the users to differentiate between students and teachers who only need to view the contents of the system, from lab personnel and IT personnel who need access to more features.

Challenges in this stage will be the configuration and implementation process itself, along with demonstrating how the enhanced inventory system is used for the students, teachers, and lab personnel ([Ghazi et al., 2023](#)).

### **Post-Survey**

After 3 months following the implementation of the enhanced system, another survey will be distributed via email. This survey will be identical to the survey used in the survey for the current inventory system. The survey will follow the same criteria, with the added restriction of having used the enhanced inventory system.

This ensures that the current system will be compared against the enhanced system during paired analysis.

### **Data Analysis Plan**

The perception-based metrics (Transparency, Satisfaction, and Reliability) will be collected with Likert-scale surveys that are given to the students, teachers, and lab personnel who actively use the laboratory. These respondents will answer this survey while the current system is in use and will answer again in three months after implementation of the enhanced system.

Transparency refers to the perceived ability of teachers, students, or lab faculty in the school to easily identify and locate specific materials or equipment from a given laboratory and determine the amount and availability of said materials and equipment. This will be measured ordinally as it is based on the perception of the user. Reliability is the perception of the users of their ability to consistently and dependably receive information as needed when using the system. As it is dependent on the user's perception, this will be measured ordinally. The satisfaction of students and employees is described as the perceived usefulness, ease of use, and overall satisfaction reported by students and employees on the inventory system, and will be measured ordinally.

## **Results and Discussion**

### **System Design, Development, and User Interface Performance**

The first objective is to design and develop a web-based laboratory inventory and borrowing system accessible to students, teachers, and laboratory personnel via standard web browsers, featuring a functional, user-friendly graphical user interface.

The system is designed to provide users with an accessible platform for checking the availability and borrowing of laboratory materials. It is to improve inventory monitoring by replacing paper-based management with a web-based platform, allowing for remote access and faster communication between the laboratory personnel and borrowers.

The development of the system should align with [Tanaman et al. \(2023\)](#), emphasizing the improvement of inventory management systems after the integration of technology into the manual process. Furthermore, other studies, such as [Ramos-Miller & Pacheco \(2023\)](#), support the need for the development of a better replacement system to reduce manual errors, allow faster communication, provide accessibility, and improve efficiency.

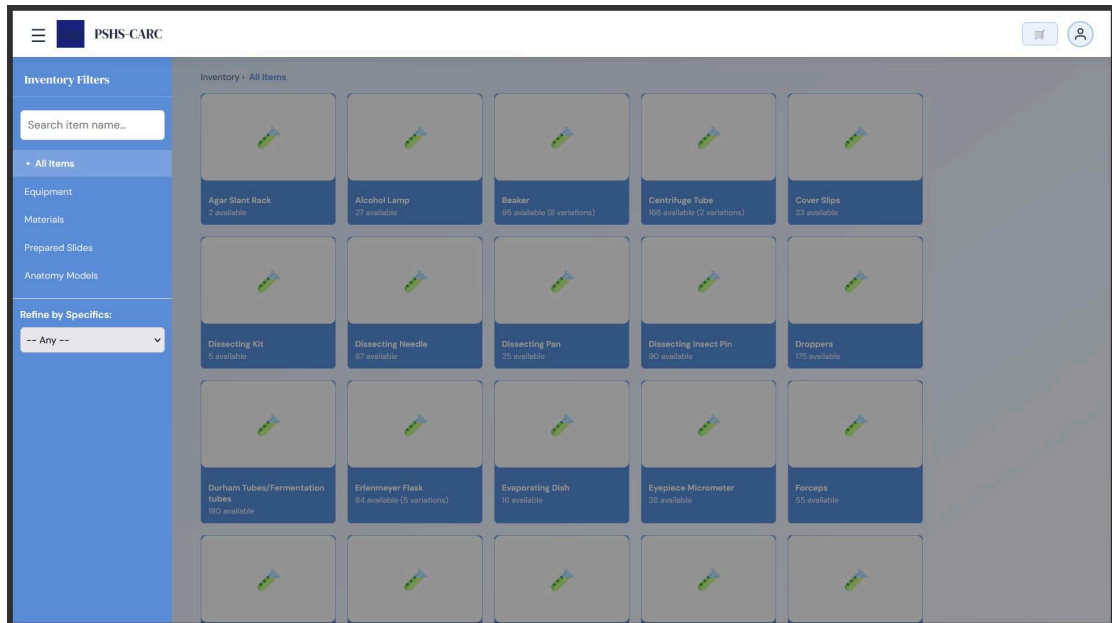


Figure 6. Home Screen of the Website with the Sidebar

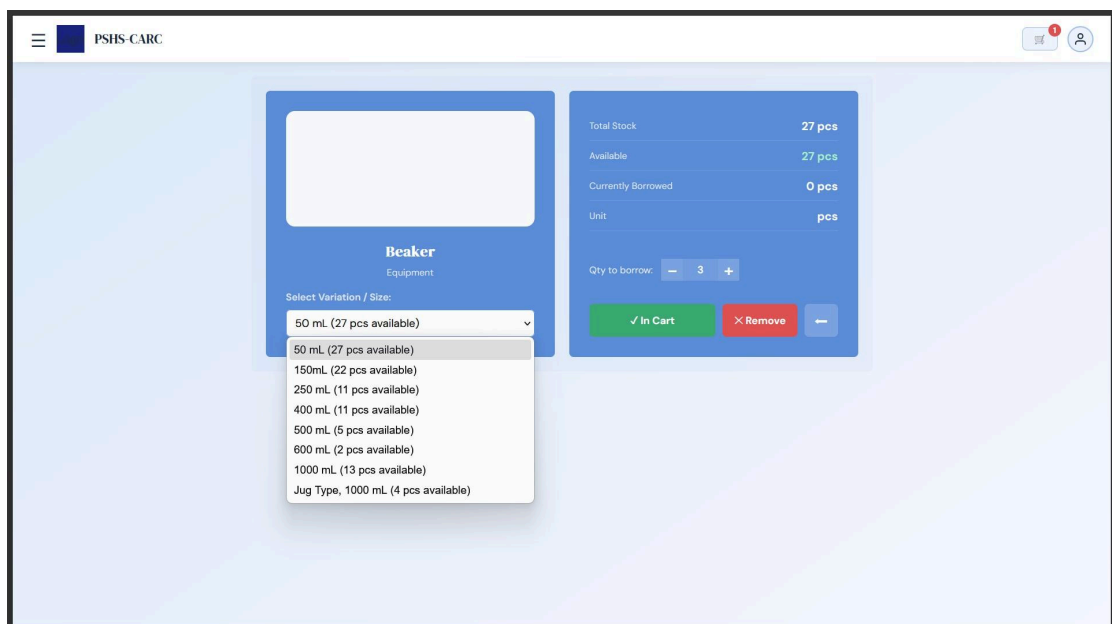


Figure 7. Equipment page

**Borrowing Request Form**  
Submit a request — lab personnel will approve before items are released.

| ITEM           | TYPE      | QTY | AVAILABLE |
|----------------|-----------|-----|-----------|
| Beaker, 400 mL | Equipment | 3   | 11 pcs    |
| Beaker, 50 mL  | Equipment | 2   | 27 pcs    |

+ Add more items

**REQUEST DETAILS**

Borrower Name \*  Section / Grade Level \*

Teacher-in-Charge \*  Role \*

Purpose / Experiment Name \*

Date & Time Needed \*  Expected Return Date \*

Additional Notes

Figure 8. Borrowing Request Form

**Request Submitted!**

Your borrowing request has been sent to the lab personnel for review. Keep your reference number for follow-up.

**BRW-375802**

**Borrower:** Lorem  
**Purpose:** BIO-Lab Experiment  
**Date needed:** 2026-05-07 12:22  
**Return by:** 2026-06-10

**Items:**  
 • Beaker, 400 mL × 3 pcs  
 • Beaker, 50 mL × 2 pcs

Status: **PENDING APPROVAL** — Lab personnel must approve before items are released.

[Make Another Request](#)

Figure 9. Confirmation of Request Page

## Effectiveness of Inventory Management Features

The second objective of the study is to implement an organized inventory management feature that classifies laboratory equipment using filters and allows users to check the availability of materials easily.

With the implementation of the inventory management components, the features are to improve the efficiency of borrowing laboratory materials. The system integrates filters that classify laboratory equipment according to item type (chemical, glassware, equipment, specimen, etc.), allowing users to easily locate specific laboratory materials. Additionally, information for each item will be provided, providing further aid for students to choose the appropriate materials to borrow.

The borrowing system's features, on the other hand, enable users to request materials on the website itself while also allowing the laboratory personnel to monitor the borrowing records. Furthermore, the availability of materials will be shown to inform users whether the materials are available and, if not, when they will be made available for borrowing.

The implementation of the system features is expected to improve efficiency in borrowing laboratory materials, reducing the time spent compared to the manual system. Additionally, it is also expected to provide faster tracking of borrowed materials, maintain accuracy, and prevent loss and mismanagement.

The effectiveness of these features will be evaluated through system testing and user feedback after data collection has been conducted, in which further interpretation will also be conducted.

### **System Evaluation**

The third objective is to assess the developed system in terms of transparency, reliability, satisfaction, and user satisfaction among students, teachers, and laboratory personnel using a questionnaire.

After four (4) months of implementation of the proposed inventory system in the biology laboratory, the post-implementation survey responses will be compared to

the pre-implementation survey responses that were previously acquired. Data for the responses will be presented in the tables below.

Table 1. Respondent data for the evaluation of the current inventory system of the biology laboratory

| Question    | Mean | Median | Standard Deviation |
|-------------|------|--------|--------------------|
| Question 1  | X.xx | X      | X.xx               |
| Question 2  | X.xx | X      | X.xx               |
| Question 3  | X.xx | X      | X.xx               |
| Question 4  | X.xx | X      | X.xx               |
| Question 5  | X.xx | X      | X.xx               |
| Question 6  | X.xx | X      | X.xx               |
| Question 7  | X.xx | X      | X.xx               |
| Question 8  | X.xx | X      | X.xx               |
| Question 9  | X.xx | X      | X.xx               |
| Question 10 | X.xx | X      | X.xx               |
| Question 11 | X.xx | X      | X.xx               |
| Question 12 | X.xx | X      | X.xx               |

Table 2. Respondent data for the evaluation of the proposed inventory system for the biology laboratory

| Question   | Mean | Median | Standard Deviation |
|------------|------|--------|--------------------|
| Question 1 | X.xx | X      | X.xx               |
| Question 2 | X.xx | X      | X.xx               |
| Question 3 | X.xx | X      | X.xx               |
| Question 4 | X.xx | X      | X.xx               |

|             |      |   |      |
|-------------|------|---|------|
| Question 5  | X.xx | X | X.xx |
| Question 6  | X.xx | X | X.xx |
| Question 7  | X.xx | X | X.xx |
| Question 8  | X.xx | X | X.xx |
| Question 9  | X.xx | X | X.xx |
| Question 10 | X.xx | X | X.xx |
| Question 11 | X.xx | X | X.xx |
| Question 12 | X.xx | X | X.xx |

Based on the comparison of the responses from the two surveys, the findings will serve as the basis for determining whether the implementation of the proposed system is relevant and useful for the users of the biology laboratory.

Depending on the results of this study, the adoption of automated inventory systems in school laboratories or laboratories in general may be supported, allowing for a more efficient and effective laboratory management system. Additionally, the system may also serve as a starting point for future expansion of the web-based inventory system for the other laboratories, as well as the school library.

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## Questionnaire 1:

The survey will have the user respond to the following statements with one of the following reactions.

- 1 - Strongly Disagree
- 2 - Disagree
- 3 - Neutral
- 4 - Agree
- 5 - Strongly Agree

| Category            | Criteria       | Indicator                                                                              | Scale<br>(1-5) |
|---------------------|----------------|----------------------------------------------------------------------------------------|----------------|
| <b>Transparency</b> | Usefulness     | I am able to easily locate specific materials or equipment using the inventory system. |                |
|                     | Content        | I can easily get the availability and quantity of laboratory items.                    |                |
|                     | Organizability | The inventory system organizes items in a way that is easy to understand.              |                |

|                     |               |                                                                                                                  |  |
|---------------------|---------------|------------------------------------------------------------------------------------------------------------------|--|
| <b>Reliability</b>  | Reliability   | I can rely on the inventory system to consistently provide information about lab resources.                      |  |
|                     | Robustness    | I can rely on the inventory system to tell me information even when multiple people are looking for information. |  |
|                     | Accuracy      | I can trust the system to reflect the actual laboratory inventory.                                               |  |
| <b>Satisfaction</b> | Utility       | The system helps me manage or access laboratory resources effectively.                                           |  |
|                     | Applicability | I find the system useful for my tasks.                                                                           |  |
|                     | Helpfulness   | The inventory system is helpful and does not impede my tasks                                                     |  |
|                     | Navigation    | The inventory system is easy to navigate                                                                         |  |

|                    |             |                                                                                                   |  |
|--------------------|-------------|---------------------------------------------------------------------------------------------------|--|
| <b>Ease of Use</b> | Usability   | I can easily use the inventory system even if it is my first time asking for information from it. |  |
|                    | Ease of use | I feel comfortable using the system independently.                                                |  |